

X Test					
	precision	recall	f1-score	support	
-1	0.36	0.48	0.42	215	
0	0.76	0.75	0.75	783	
1	0.72	0.64	0.68	463	
accuracy			0.67	1461	
macro avg	0.62	0.62	0.62	1461	
weighted avg	0.69	0.67	0.68	1461	
X Train					
	precision	recall	f1-score	support	
-1	0.56	0.77	0.65	645	
0	0.87	0.81	0.84	2347	
1	0.85	0.79	0.82	1389	
accuracy			0.80	4381	
macro avg	0.76	0.79	0.77	4381	
weighted avg	0.81	0.80	0.80	4381	

Figure 3: Results from testing the trained MNB algorithm

Step 9: Let us evaluate the performance of the trained MNB algorithm using the test data:

```
nb_pred = nb_model.predict(X_test_count)
nb_train_pred = nb_model.predict(X_train_count)
print("X Test")
print(classification_report(y_test,nb_pred))
print("X Train")
print(classification_report(y_train,nb_train_pred))
```

The output of the code above has been shown in Figure 3. We have successfully trained and tested the Multinomial Naïve Bayes algorithm on the data set, which can now predict the sentiment of a statement from financial news with 80 per cent accuracy.

Step 10: We can visualise these results in the form of a heatmap for more clarity:

```
plt.figure(figsize=(8,8))
sns.heatmap(confusion_matrix(y_test,nb_pred),annot = True,fmt = "d")
```

Figure 4 shows the results of the MNB algorithm in the form of a heatmap.

Step 11: We can also visualise these results using a precision-recall curve:

```
from yellowbrick.classifier import PrecisionRecallCurve
viz = PrecisionRecallCurve(MultinomialNB(),
                           classes=nb_model.classes_,
                           per_class=True,
```

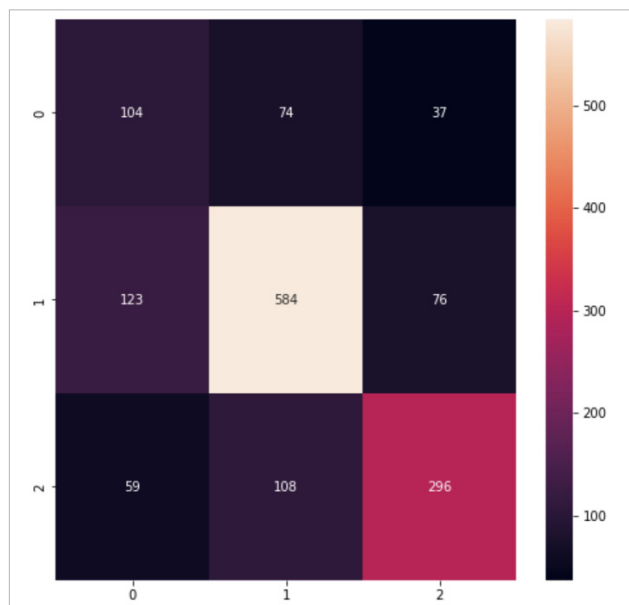


Figure 4: MNB algorithm accuracy for data set

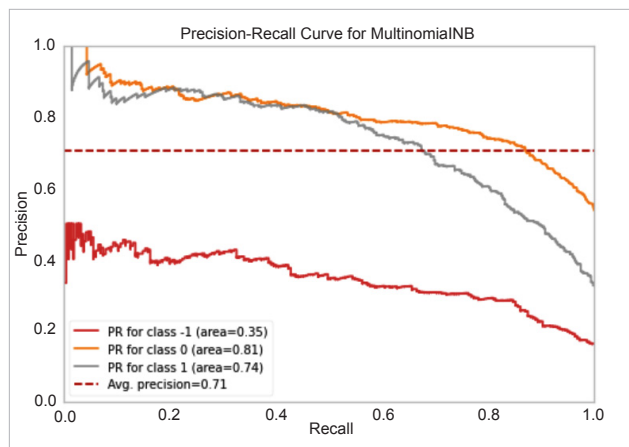


Figure 5: Precision-recall curve of the MNB algorithm

```
cmap="Set1")
viz.fit(X_train_count,y_train)
viz.score(X_test_count, y_test)
viz.show();
```

Figure 5 shows the precision-recall curve of the MNB algorithm. 

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